

Smart Disaster Resource Exchange

Team - The Trio

Aarya Gawade

Udayagiri Narayana

Pranav Vijayakumar



Problem Statement

During crises, critical resources frequently exist nearby but are invisible: hospitals, police, NGOs, fire crews and volunteers work from siloed systems. Emergency coordinators waste time calling around, verifying availability, and tracking requests manually. Citizens lack a single place to request help or see status.

Coordination gap

Fragmented systems across agencies cause delays and duplication of effort.

Slow discovery

Finding available resources requires time-consuming manual outreach.

Low transparency

Citizens can't track requests in real time or know where help stands.

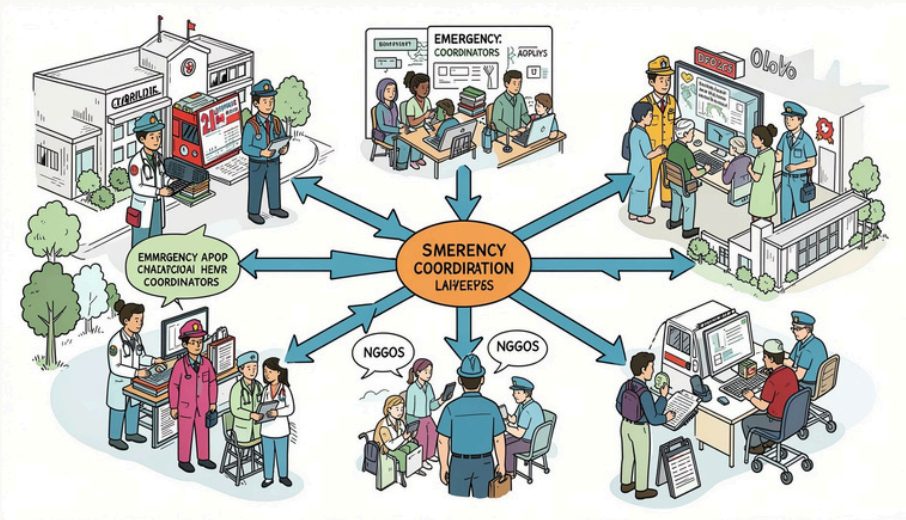
Opportunity

Create a centralized exchange to publish, discover, request, and allocate resources fast.

Solution Overview & System Architecture

How it works (workflow)

1. Citizen submits a request (location, need, urgency).
2. Coordinator validates and prioritizes using the dashboard.
3. System matches nearby resource owners with availability.
4. Owner accepts/rejects; system tracks allocations.
5. Citizen receives real-time status updates and ETA.



Tech stack & architecture highlights

- **Frontend:** React.js — role-aware views for citizens, coordinators, owners.
- **Backend:** Node.js + Express with REST + GraphQL endpoints.
- **DB:** PostgreSQL (Supabase) for geo-indexed inventories and audit logs.
- **Authentication:** Cisco Identity Services Engine (ISE) for secure, role-based authentication and access control.
- **Monitoring:** Cisco AppDynamics for real-time backend performance monitoring and application health.
- **Network Visibility:** Cisco ThousandEyes for end-to-end network monitoring and connectivity insights.



Real-time alerts

Push notifications and SMS for request updates and acceptance.



Geo-matching

Location-based search prioritizes nearby resources and reduces transit time.



Secure & auditable

Role-based access and an immutable activity log for accountability.

MVP Features, Impact & Demo Flow

Expected impact



Role-based secure login

Distinct flows and permissions for citizens, coordinators, and resource owners.



Coordinator dashboard

Prioritize, assign, and monitor requests with a single pane of glass.



Inventory & approvals

Publish available resources; owners approve allocations with one tap.

- Faster discovery & allocation — less time triaging, more time delivering aid.
- Reduced cross-agency phone traffic and duplication.
- Higher utilization of existing assets across NGOs, hospitals, and government.
- Transparency for citizens with live status and ETA updates.
- Scalable to connect with regional/national emergency systems.

Demo flow (MVP)

Citizen → Submit request (photo, location, urgency) → Coordinator validates & assigns → Resource owner receives request and accepts → System records allocation; dispatch begins → Citizen gets live status updates and confirmation on delivery.

❏ Architecture diagram link: [Online FlowChart & Diagrams Editor - Mermaid Live Editor](#)

Github link: [narayana-srimanth/cisco-project](#)